

الجمهورية الجزائرية الديمقراطية الشعبية
 People's Democratic Republic of Algeria
 وزارة التعليم العالي والبحث العلمي
 المؤسسة الوطنية للأمن السيبراني

Ministry of Higher Education and Scientific Research
 National School of Cyber Security

NSCS
 المدرسة الوطنية للأمن السيبراني
 NATIONAL SCHOOL OF CYBERSECURITY

MODULE'S TEACHER: Dr. Hamza Moufek
 DATE: 09 / 04 / 2026
 DURATION: 1h30
 Note: No documents are allowed.

LEVEL: 1st Year Basic Training
 SECTION / GROUP: A & B
 MODULE: Algebra 2
 FULL NAME:

Midterm Test

Exercise 1 : (4 points)

Let

$$P = (X - 5)^2 \quad \text{and} \quad M = \begin{pmatrix} 6 & 1 & 1 \\ 1 & 6 & 1 \\ -2 & -2 & 3 \end{pmatrix}.$$

1. For every $n \in \mathbb{N}$, determine the remainder of the Euclidean division of X^n by P .
2. Compute $P(M)$ and deduce M^n in terms of n , for every $n \in \mathbb{N}$.

Exercise 2 : (6 points)

Let $a \in \mathbb{R}$. In the vector space $\mathbb{R}^3$, define the vectors $u_a = (a, 1, 1)$ and $v = (1, -2, -2)$.
 Let $E_a = \text{Span}\{u_a, v\}$.

1. Determine, depending on the value of a , the dimension of E_a .

We define $E = \{(x, y, z) \in \mathbb{R}^3 \mid y - z = 0\}$.

2. Show that E is a subspace of $\mathbb{R}^3$.
 Determine a basis of E , and specify its dimension.
3. Show that $E = E_1$.

Exercise 3 : (10 points)

Consider the matrices

$$A = \begin{pmatrix} 16 & 4 & -4 \\ -18 & -4 & 5 \\ 30 & 8 & -7 \end{pmatrix} \quad \text{and} \quad P = \begin{pmatrix} 1 & 0 & -1 \\ -2 & 1 & 1 \\ 2 & 1 & -2 \end{pmatrix}.$$

Part I : Diagonalization

1. Show that the matrix P is invertible and compute its inverse.
2. Let $D = P^{-1}AP$. Show that $D = \text{Diag}(0, 1, 4)$.
3. (a) Compute the powers of the matrix D .
 (b) Deduce the powers of the matrix A .
4. Show that for every matrix $M \in \mathcal{M}_3(\mathbb{R})$, the matrices M and D commute if and only if the matrix M is diagonal.

Part II : Square roots of the matrix

In this part, we aim to solve the equation $M^2 = A$, where the unknown matrix is $M \in \mathcal{M}_3(\mathbb{R})$.

5. Let $M \in \mathcal{M}_3(\mathbb{R})$ be such that $M^2 = A$. Define $B = P^{-1}MP$.

(a) Show that $B^2 = D$.

(b) Deduce that $BD = DB$.

(c) Deduce that the matrix B can be written as

$$B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \gamma & 0 \\ 0 & 0 & 2\delta \end{pmatrix}, \quad \text{with } (\gamma, \delta) \in \{-1, 1\}^2.$$

6. Conversely, let $B \in \mathcal{M}_3(\mathbb{R})$ be such that $B^2 = D$. Define $M = PBP^{-1}$.

What can be said about the matrix M ?

7. Solve the equation $M^2 = A$, where the unknown is $M \in \mathcal{M}_3(\mathbb{R})$.